

Sporting Ethos — Milestone 2: Functional MVP & Validated Workflow

GigPoint Hackathon 2026 · Track 4 (Healthcare Tech) Deliverable: Functional MVP/Prototype + validated end-to-end workflow

1. What was built

A complete, working clinic operations system that eliminates check-in uncertainty and extends into charting, pharmacy billing, and analytics — one responsive web app + a small AI backend, live on Supabase.

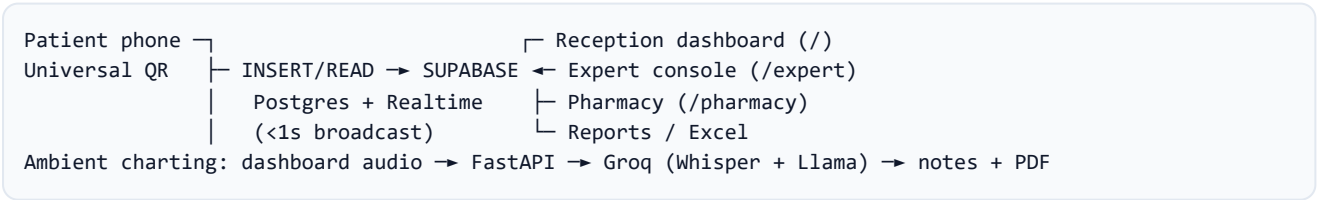
Core (the brief's problem — solved):

- **Universal QR check-in** — one code for everyone; sequential auto appointment IDs (APT-0001...).
- **Sub-second live queue** for the expert/reception (Supabase Realtime, no refresh).
- **Multi-sensory arrival confirmation** — on-screen highlight + chime + spoken "<name> has arrived."
- **Verifiable receipt** — SHA-256 hash, UUID, timestamp.
- **Full status lifecycle** — waiting → in_consult → done, plus left (LWBS) / no-show / paused.
- **Emergency priority** — jumps the queue; all positions/ETAs recalculate on every phone.
- **Call-next** rings the patient's phone (chime + voice, in their language).

High-impact extensions:

- **Ambient AI charting** (FastAPI + Groq Whisper + Llama 3.3 70B): consultation audio → transcript → structured notes (Symptoms/Prescriptions/Actions) → PDF → patient visit summary.
- **Pharmacy** (/pharmacy): dispense by appointment ID → prescriptions auto-matched to inventory → bill + payment → **PDF invoice** + stock decrement; plus inventory management.
- **Reports** — KPIs, demographic donuts/bars, arrivals trend, **top reasons/conditions (from consultations)**, **pharmacy revenue**, and one-click **Excel export**.
- **Expert Console** (/expert) + **reception intercom** (two-way call bell).
- **Reception " + Add patient"** (desk walk-in registration).
- **3 languages** (English/Hindi/Telugu) + **accessibility mode** (large text + voice guide).

2. Architecture (brief)



Stack: React (Vite) + Tailwind · Supabase (Postgres + Realtime) · FastAPI + Groq · Web SpeechSynthesis (voice) · Web Crypto (SHA-256) · jsPDF (receipt/bill) · SheetJS (Excel).

3. End-to-end workflow — validated

#	Flow	Result
1	Scan QR → check in (name/age/gender)	Sequential APT-#### assigned; row in Postgres
2	Dashboard receives arrival	Appears in <1s ("Confirmed in <1s") + highlight + chime + voice

#	Flow	Result
3	Call next	Patient phone shows "It's your turn" + chime/voice; status → in_consult
4	Expert records consultation	Groq returns real structured notes (verified live)
5	Mark done	Notes saved; patient sees visit summary
6	Pharmacy: enter APT id	Prescriptions auto-matched to inventory → bill → payment → PDF, stock ↓
7	Reports	Live demographics, reasons, and pharmacy revenue; Excel export
8	Emergency flag	Patient jumps to #1; all ETAs update

All verified against the **live Supabase database**; production build passes (0 errors).

4. Roles & access

Role	URL	Access
Reception	/	open (kiosk)
Expert	/expert	code ethos
Pharmacy (counter + inventory)	/pharmacy	code pharmacy
Patient	/checkin	via QR

5. How to run

```
# Frontend
npm install && npm run dev      # open the Network URL (192.168.x.x) for phones

# Ambient charting backend
cd backend && pip install -r requirements.txt
# add GROQ_API_KEY to backend/.env (free: console.groq.com)
python -m uvicorn main:app --port 8000
```

Supabase is connected via `.env` ; the dashboard badge shows **"Live"**.